1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with favorable immune architecture still fail to achieve better clinical outcomes?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), the prognostic impact of a high stromal-to-tumor ratio (TSR) on overall survival (OS) is contingent on tumor genotype, specifically MSH6 status and DPYD expression. TSR-high tumors that are MSH6-deficient (loss of function or MSI-H phenotype) and exhibit high DPYD expression will create a stromal milieu that promotes immunosuppression and metabolic adaptation, enabling immune escape and reduced sensitivity to perioperative therapy, resulting in poorer OS. Conversely, TSR-high tumors with intact MSH6 or low DPYD expression will show a less immunosuppressive stromal environment and greater chemosensitivity, yielding OS comparable to TSR-low tumors. This defines a three-way interaction (TSR × MSH6 × DPYD) that remains significant after adjusting for standard clinical covariates (age, sex, stage, nodal status, performance status) and treatment variables (neoadjuvant/adjuvant chemotherapy, surgical margins). The hypothesis predicts: (1) a significant three-way interaction term predicting OS across cohorts; (2) within TSR-high tumors, the worst outcome in the MSH6-deficient, DPYD-high subgroup; (3) correlative immuno- and metabolic signatures (e.g., reduced CD8+ T cells, increased Tregs/M2 macrophages, altered pyrimidine metabolism gene programs) accompanying the genotype–stroma–host context; and (4) mechanistic recapitulation in model systems where manipulating MSH6 and DPYD alters stromal-immune crosstalk and chemoresistance. These relationships should be detectable in retrospective cohorts and validate in prospective samples, with complementary in vitro and in vivo experiments guiding causal interpretation.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MSH6, DPYD
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6, DPYD) to satisfy the focusing rule and produce a sharp mechanistic question; SARIFA (the stromal–adipocyte concept), ALB/GLB, HRR, SDC2/TFPI2, and PTGDS were placed in secondary_modifiers rather than the core claim to avoid overfitting the primary hypothesis and to preserve testability.

DPYD: DPYD) to satisfy the focusing rule and produce a sharp mechanistic question; SARIFA (the stromal–adipocyte concept), ALB/GLB, HRR, SDC2/TFPI2, and PTGDS were placed in secondary_modifiers rather than the core claim to avoid overfitting the primary hypothesis and to preserve testability.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SARIFA	Supporting modifier from knowledge graph
Secondary	ALB/GLB	Supporting modifier from knowledge graph
Secondary	HRR	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: TSR as a stromal biomarker captures ECM remodeling and immune cell localization that influence prognosis but is not universally predictive, suggesting context dependence;

Prediction 2: MSH6 deficiency (a component of the MMR pathway) is linked to increased mutational burden and neoantigen load yet can be associated with immunosuppressive microenvironments and altered DNA repair–linked metabolism; DPYD encodes dihydropyrimidine dehydrogenase, a key regulator of pyrimidine catabolism that can modulate nucleotide pools and influence sensitivity to pyrimidine-targeting therapies and metabolic adaptation;

Prediction 3: the host systemic state (nutritional/inflammatory status) and epigenetic/metabolic signaling in the tumor–stroma axis can further modulate outcomes, supporting a genotype × stroma × host context framework.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SARIFA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ALB/GLB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.